

Table 1

Brick	This is the basic unit of storage, represented by an export directory on a machine
Server	This is the machine that hosts the actual file system in which the data will be stored. A node is a server capable of hosting bricks.
Client	This is the machine that mounts the volume (this may or may not be the server).
Trusted storage pool	A storage pool is a trusted network of storage servers. You can dynamically add or remove nodes from the pool. Only nodes that are in a trusted storage pool can participate in volume creation.
Volume	A volume is a logical collection of bricks. Most of the Gluster management operations happen on the volume. Multiple volumes can be hosted on the same node.
Translator	These are the building blocks of GlusterFS. All functions are implemented as translators. They are stacked together to achieve the desired functionality.
Volfile	.vol files are configuration files used by the <i>glusterfs</i> process. A set of translators stacked to achieve a particular functionality is stored in a volfile, which is usually located at <code>/var/lib/glusterd/vols/volume-name/</code>

peers. For example, if nodes A and B are peers, and if node A adds node C as a peer, then nodes A, B and C all become peers and can participate in the creation of volumes. We can also remove nodes from the trusted storage pool. To create a volume we can use bricks from any number of nodes, provided they are in the same trusted storage pool. On one node, we can create multiple bricks and hence can host multiple volumes.

As a part of volume creation *glusterd* will generate client and server *volfiles*. A *volfile* is a configuration file that has a collection of translators, where each translator has a specific function to perform and together implement the desired functionality. These files are stacked in a hierarchical structure. A translator receives data from its parent translator, performs the necessary operations and then passes the data down to its child translator in the hierarchy.

Winding fop via the client-server graph

On mounting a volume on a client machine, a *glusterfs* client process is spawned, which interacts with the *glusterd* process and gets a configuration file or the client *volfile*. *Glusterd* also sends a list of ports on which the client can talk to each brick.

When we issue a file operation or *fop*, VFS hands the request to the FUSE module which, in turn, hands the request to the GlusterFS client process. The GlusterFS client processes the request by winding the call to the first translator in the client *volfile*, which performs a specific operation and winds the call to the next translator. The last

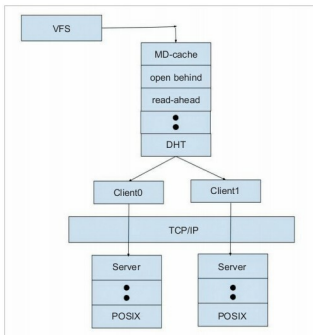


Figure 1: The client-server graph

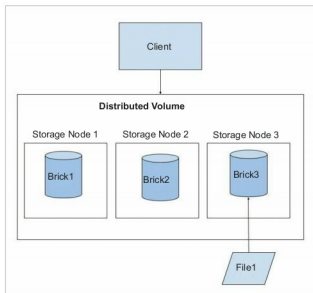


Figure 2: Distributed GlusterFS volume

translator in the client *volfile* is the protocol client, which has the information regarding the server side bricks and is responsible for sending the request to the server (each brick) over the network. On the server stack, the first translator loaded in the protocol server is responsible for getting the package. Depending on the options set, there might be a few more translators below it. Finally, the request reaches the POSIX file system.

Types of volumes

The Gluster file system supports different types of volumes,